

DATA VISUALISATION PROJECT 2026

Airbnb & the Milan Rental Market

Exploring the spatial and temporal correlation between
short-term rentals and long-term rent prices in Milan (2018-2024)

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GitHub Repository	github.com/Mturco3/Data-Visualization-2026
Interactive Dashboard	Tableau Public - Milan Housing Dynamics
Project Website	mturco3.github.io/Data-Visualization-2026
Airbnb Data	insideairbnb.com/get-the-data
OMI Milan Data	dati.comune.milano.it - OMI

Analysis performed for educational purposes only. All data is publicly available.

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1. Introduction

Over the past decade, platforms like Airbnb have transformed how people travel and, crucially, how cities allocate their housing stock. In Milan, Italy's economic capital and one of Europe's most visited cities, the number of active short-term rental listings has grown dramatically since 2018, raising a pressing question for residents and policymakers alike:

How has the rise of short-term rentals reshaped Milan's urban housing landscape between 2018 and 2024, and is there a measurable spatial and temporal correlation with the long-term rental market?

This project uses Milan as a case study of a broader European phenomenon. We combine two public datasets, Airbnb listing data and OMI official rent estimates from the Italian Revenue Agency, to map where short-term rental pressure is highest and whether those same areas show rising long-term rents.

Honest scope. We measure *market pressure* (correlation between Airbnb growth and rent levels), not direct displacement. Measuring displacement would require eviction or registry data we do not have. The distinction between pressure and displacement is itself an interpretive choice, and we make it explicit in Section 5.

Target audience. Urban planners, students, local residents, and policy observers who need clear spatial and temporal evidence rather than purely technical metrics.

2. Data Sources

2.1. Inside Airbnb - Milan

Listing-level snapshot from Inside Airbnb (2025), with the embedded review history used as a temporal proxy to reconstruct activity from 2018 to 2024. Source: insideairbnb.com/get-the-data.

- **Reviews dataset:** Review-level records (listing_id and review date) are used to estimate yearly Airbnb activity when full historical availability is not available.
- **Granularity:** Individual listing, aggregated by neighbourhood and OMI zone.
- **Key fields:** host_id, room_type, price, reviews_per_month, calculated_host_listings_count, license (CIN), listing_id, review date.
- **Limitations:** Scraped data; availability calendars are unreliable; no direct revenue figures; the review-based activity proxy underestimates listings with few reviews.

2.2. OMI - Agenzia delle Entrate

Official long-term rent estimates published by the Italian Revenue Agency, by OMI zone and semester (2018 H1 to 2024 H2). Source: dati.comune.milano.it - OMI.

- **Granularity:** OMI zone, mapped to Airbnb neighbourhoods via a custom crosswalk.
- **Key fields:** zona_omi, typology, rent_min, rent_max, semester.
- **Limitations:** OMI publishes estimated ranges, not actual transaction prices; OMI zone boundaries do not align with Airbnb neighbourhoods; some zones were merged or redrawn during the period.

2.3. Custom Neighbourhood-Zone Crosswalk

OMI zones and Airbnb neighbourhoods use different boundary systems. We built a curated crosswalk that links each Airbnb neighbourhood to one or more OMI zones, which lets us attach official rent context to every listing in the snapshot.

3. Interactive Dashboard

The Tableau dashboard summarises Milan's Airbnb pressure and housing-market context between 2019 and 2024. It combines citywide KPIs, a neighbourhood map, Airbnb review growth, OMI rent growth, residential typology rent levels, and the relationship between average Airbnb price and average rent level.

Access the dashboard:

public.tableau.com/.../Milan_Housing_Dynamics/Dashboard1

The dashboard offers four interactive views:

- **Map:** select a Milan zone to inspect local Airbnb and rent-market indicators.
- **KPI strip:** average Airbnb price, OMI level, total listings, reviews, and distinct hosts.
- **Trend views:** Airbnb review growth rate and residential rent-level growth rate from 2019 to 2024.
- **Comparison views:** OMI rent by residential typology and Airbnb price versus rent level.

4. Exploratory Analysis and Key Visualisations

4.1. Overview: Current Offer and Neighbourhood Concentration

The current listing stock is dominated by *entire homes*, which account for approximately 87.9 % of the snapshot. Five neighbourhoods (Buenos Aires-Venezia, Duomo, Sarpi, Centrale, and Loreto) concentrate roughly 28 % of all listings, indicating strong spatial clustering in central and tourist-attractive areas.

4.2. Minimum-Stay Requirements

An analysis of the `minimum_nights` field reveals how hosts position their listings on the market. Most listings fall in the short-stay category (1 to 6 nights), confirming a predominantly tourist-oriented supply. A non-trivial share of listings requires 30 or more nights, signalling segments that operate closer to the medium-term rental market.

4.3. Temporal Dynamics: 2018 to 2024

Review activity is used as a proxy for listing activity over time. Key observations:

- Review activity fell by approximately 67.5 % from 2019 to 2020, reflecting the impact of the COVID-19 pandemic on tourism.
- Recovery was rapid and strong. By 2024, the yearly review count was approximately 329 % above the 2019 level.

- Official OMI rents moved more sluggishly: they declined modestly in 2020 to 2021 and then recovered, but without matching the dramatic rebound in Airbnb activity.

4.4. 3D Isometric Map: “The Invasion”

A Three.js isometric visualisation encodes two dimensions simultaneously: palace height represents the OMI midpoint rent (€/m²/month) for each neighbourhood, while floating balloons above each palace represent review-inferred active listings (one balloon per 80 reviewed listings). Selecting any year from 2018 to 2024 animates the change in both dimensions, making spatial inequality and temporal growth immediately visible.

The visualisation is accessible on the project website: mturco3.github.io/Data-Visualization-2026.

5. White Hat / Black Hat

The same dataset, two radically different framings. Both are technically accurate. Only one is honest.

5.1. White Hat: Transparent and Constructive View

The white hat figure uses the full overlapping period from 2018 to 2024 and indexes both series to the same baseline (2018 = 100). The reader can see that Airbnb review activity rebounds much faster than official rents, but the figure does not imply direct causation. The design choices (shared baseline, full time window, single axis, and neutral annotation) give the reader a fair chance to judge how similar or dissimilar the trajectories really are.

5.2. Black Hat: Misleading Framing

The black hat figure uses the same underlying citywide series but applies several manipulative design decisions:

1. Only the post-2021 rebound is shown, erasing the pandemic break and making the recovery look like a seamless acceleration.
2. Dual vertical axes with different scales amplify the visual alignment between two variables that live on completely different magnitudes.
3. The rent axis is truncated to exaggerate variation.
4. The title turns a visual coincidence into a causal slogan: *“Airbnb explodes and Milan rents follow immediately”*.

The lesson is useful beyond this project: ethical problems in visualisation often come from emphasis and framing rather than fabricated data. Every data point can be real, yet the reader is still pushed toward a stronger conclusion than the evidence supports.

6. Key Findings

1. **Concentration is real but not citywide uniform.** The five largest neighbourhoods account for about 28 % of all listings, with Buenos Aires-Venezia and Duomo standing out most clearly.

2. **Current Airbnb presence aligns with expensive rent contexts.** The correlation between listing stock and linked 2024 OMI midpoint rent is positive ($r \approx 0.62$). This is a cross-sectional association, not causal proof.
3. **COVID was a sharp interruption, not a reset.** Review activity fell by about 67.5 % from 2019 to 2020, then rebounded strongly. By 2024 the yearly review count was roughly 329 % above the 2019 level.
4. **The stock is heavily commercialised.** Entire homes represent about 87.9 % of the snapshot, and hosts with at least five listings account for about 45.1 %.
5. **Pressure does not automatically imply displacement.** The data show a meaningful overlap between Airbnb concentration and high rent context, but not direct evidence of displacement. That would require eviction or registry data.

7. Reproducibility and Project Resources

Resource	Link
GitHub repository	github.com/Mturco3/Data-Visualization-2026
Project website	mturco3.github.io/Data-Visualization-2026
Tableau dashboard	public.tableau.com/.../Milan_Housing_Dynamics
Airbnb source data	insideairbnb.com/get-the-data
OMI Milan data	dati.comune.milano.it/dataset?tags=OMI

Table 1: Project links and data sources.

All analysis is performed in Python (pandas, matplotlib, seaborn) within Jupyter notebooks located in the `src/` directory. The D3.js and Three.js interactive components are in the `docs/` directory and are served via GitHub Pages at the project website above.

AI Tools Disclosure. Parts of this project, including code assistance, text drafting, and visualisation refinement, were developed with the support of AI tools (Claude, Anthropic). All analytical decisions, interpretations, and conclusions remain the authors' own. Data sources are publicly available and independently verified.

Disclaimer: Analysis performed for educational purposes only. All data is publicly available.